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Kinematic correlates of kinetic outcomes associated with running-related injury

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Abstract

High magnitudes and rates of loading have been implicated in the etiology of running-related injuries (RRI). Knowledge of kinematic variables that are predictive of kinetic outcomes could inform clinic-based gait retraining programs. Healthy novice female runners ran on a treadmill while 3D biomechanical data were collected. Kinetic outcomes consisted of vertical impact transient (VIT), average vertical loading rate (AVLR), instantaneous vertical loading rate (IVLR), and peak braking force (PBF). Kinematic outcomes included step length (SL); hip flexion angle at initial contact (HFA); horizontal distance from heel to centre of mass at initial contact (HCOM); shank angle at initial contact (SA); and foot strike angle (FSA). Stepwise multiple linear regression was used to evaluate the amount of variance in kinetic outcomes explained by kinematic outcomes. A moderate amount of variance in kinetic outcomes (VIT=46%; AVLR=37%; IVLR=49%; PBF=54%) was explained by several discrete kinematic variables—predominantly speed, HCOM, FSA, and SL. HFA and SA did not contribute to any models. Decreasing SL and transitioning from a rearfoot strike may reduce kinetic risk factors for RRI. In contrast, clinical strategies such as modifying SA and HFA would not appear to contribute significantly to the variance of kinetic outcomes after accounting for other variables.

Keywords: running, clinical biomechanics, gait analysis, kinematics, kinetics

Word Count: 4730

Introduction

Running is one of the most popular activities worldwide, with numerous benefits in the prevention of obesity, hypertension, type 2 diabetes, osteoarthritis, respiratory disease, cancer, and disability.¹ Even in low doses, running is associated with a substantial reduction in cardiovascular and all-cause mortality.¹ However, injury rates are high and have not decreased appreciably in the decades since monitoring began.²⁻⁴ While the etiology of running injuries is multifactorial,⁵ biomechanical factors—primarily kinetic factors associated with the loading phase of running—have been linked to the development of running-related injury.⁶⁻¹² For instance, a high vertical loading rate (whether measured as an average or instantaneous variable) has been cited as a risk factor for tibial stress fracture in two systematic reviews.^{6,11} Another recent study¹³ reported that peak braking force may also be an important risk factor for running-related injury, with higher braking forces resulting in risk of injury five to eight times that of lower braking forces.

Recently, running gait retraining has become a popular mode of treatment for running injuries, with promising results.^{14,15} Several studies have shown the ability to lower vertical loading rates through a gait retraining intervention with real-time biofeedback of vertical ground reaction force or tibial acceleration.¹⁵⁻¹⁷ To date, the most effective interventions have relied on lab-based feedback of kinetic metrics.¹⁸ However, in a clinical setting, it is not always feasible to provide real-time feedback of ground reaction force variables due to constraints on equipment, costs, and analytic methods. Fortunately, many kinematic variables can be measured and modified in a clinical setting using 3D motion capture or even 2D video.^{19,20} To better design and implement gait retraining interventions, it is important to understand which kinematic variables contribute to kinetic risk factors of running-related injury.

The posture of the lower extremity at initial contact and during the loading phase has been reported to influence the magnitude and rate of loading.²¹⁻²³ In a step rate intervention, Heiderscheit et al²¹ found that decreased step frequency—or greater step length—resulted in a higher braking force and active peak. The increased step length was accompanied by an increase in the horizontal distance between the heel and centre of mass at initial contact. Further, Mercer et al²⁴ reported that changes in step length, but not step frequency, affected shock attenuation in runners. They posited that shock attenuation, by way of tibial acceleration magnitude, was affected by changes in step length due to alterations in lower extremity posture and compliance at initial contact. Lieberman et al²² noted that increases in step length can be achieved either by a longer flight phase or by landing with a more extended lower extremity with a greater heel to centre of mass distance at initial contact (often referred to as “overstriding”).

Furthermore, the unique combination of hip and knee flexion at initial contact can affect the appearance of this overstride, with greater knee extension and hip flexion—resulting in an increase in shank angle and horizontal distance from heel to centre of mass—producing greater braking force and impact peak magnitude.²² Heiderscheit et al²¹ observed a decrease in knee flexion and an increase in hip flexion at initial contact with increasing step length, but did not report the relationship between these kinematic variables and the kinetic outcomes. In a clinical context, increased step length, heel to centre of mass distance, hip flexion angle at initial contact, and shank angle are all believed to be measures of overstriding. Relating these measures and their contribution to kinetic outcomes associated with running-related injury, therefore, is an important step to be able to inform clinical guidelines for gait retraining. While increased step length²⁵ and heel to centre of mass distance^{23,25} have been linked previously to higher vertical loading rates and braking impulse, to our knowledge no previous study has investigated the contribution of hip

flexion angle or shank angle at initial contact to kinetic outcomes associated with running-related injury risk.

Foot strike angle at initial contact has also been reported to change significantly with greater changes ($\pm 10\%$) in step length, with a more dorsiflexed ankle at initial contact at longer step lengths.²¹ In a study by Wille et al.,²³ foot strike angle at initial contact was present in all models used to predict kinetic outcomes, indicating its important influence on the loading phase of running. Vertical ground reaction force measures have been consistently reported to be lower in runners with lower foot strike angles.²⁶⁻²⁸ Given the association between foot strike angle and loading, it is likely that foot strike angle could explain a significant amount of the variance in kinetic outcomes.

Overstriding has been suggested to be important in running-related injury pathogenesis.¹⁴ Therefore, knowledge of specific kinematic variables that relate to overstriding—and are predictive of kinetic risk factors—could inform clinic-based gait retraining programs aimed at reducing risk factors of running-related injury. The purpose of this study, therefore, was to determine the association between clinically-measurable kinematic variables and kinetic running-related injury risk factors. Specifically, we investigated the correlation between five kinematic variables (step length, hip flexion angle at initial contact, horizontal distance between the heel and centre of mass, shank angle at initial contact, and foot strike angle) and four kinetic outcomes (peak braking force, average vertical loading rate, instantaneous vertical loading rate, and vertical impact transient). We also examined the contribution of these kinematic variables to the variance in kinetic outcomes. We hypothesized that a greater step length, landing with a more flexed hip, a greater horizontal distance between the heel and centre of mass, a greater shank angle at initial contact, and a more rearfoot strike angle would be associated with a greater peak braking force,

vertical impact transient, average vertical loading rate, and instantaneous vertical loading rate during running. We also hypothesized that these kinematic variables would significantly contribute to all kinetic outcome models.

Methods

Participants: Data presented in this study are from the baseline testing session in a previous longitudinal study examining running-related injury incidence over a 15-week running program.¹³ A sample of female recreational runners between the ages of 18 and 60, and running for at least three months was recruited from the local running community via flyers and social media posts. Participants were excluded if they had experienced (1) a lower extremity injury in the previous three months, (2) any history of lower limb surgery, or (3) any current low back or lower extremity pain while running. Written consent was obtained from all participants and ethics approval was granted from the institutional Clinical Research Ethics Board.

Data collection: Demographics and a detailed training and injury history were collected for each participant prior to undergoing a biomechanical running analysis on an instrumented force treadmill (Treadmetrix LLC, Park City, USA). Sixty-one reflective markers were affixed to each participant prior to testing, and a static calibration trial was initially collected to form a musculoskeletal model based on MacLean et al²⁹ (Figure 1) using a six-camera motion analysis system (Qualisys Motion Capture Corp., Gothenburg, Sweden). Nineteen static/calibration markers (vertex of the head, medial elbow epicondyles, anterior superior iliac spines, greater trochanters, medial/lateral femoral condyles, medial/lateral malleoli, and first and fifth metatarsal heads) were removed following the static trial and the 42 remaining markers (five on the head, C7 spinous process, acromioclavicular joints, mid-triceps, lateral epicondyles, distal radius/ulna, posterior superior iliac spines, iliac crests, clusters of four on the thigh and shank, and a triad on

the heel) were by definition tracking and calibration markers as they were on for both static and dynamic trials. The ankle joint centre was identified as the midpoint between the two ankle malleoli markers; the knee joint centre was identified as the midpoint between the two femoral condyle markers; and the hip joint centre was identified using the method of Bell and Brand.³⁰

Participants ran in their usual running shoes at a self-selected speed representative of a moderate intensity run (Borg Rating of Perceived Exertion = 13, indicative of “somewhat hard”).³¹ Three-dimensional kinetic (sampled at 2400 Hz) and kinematic (240 Hz) data were collected from three consecutive captures of 15 seconds duration each during the last minute of a single continuous running bout of 3-5 minutes.

Data analysis: Biomechanical data were analyzed using Visual 3D software (C-Motion, Inc., Germantown, USA). Kinetic and kinematic data were low-pass filtered using a critically-damped digital filter at a cutoff frequency of 20 Hz and 8 Hz, respectively.³² Initial contact and toe-off events were identified by a vertical ground reaction force threshold of 50 N and were used to determine the stance and swing phases of the gait cycle. Kinetic variables were normalized to body weight.

The study limb was chosen based on the criteria described in Napier et al.¹³ All biomechanical outcome measures were determined from the first 10 consecutive stance phases for the study limb within each of the three captures (n=30 total stance phases per participant). Kinetic outcomes were selected based on their established importance in the running-related injury literature, and consisted of vertical impact transient,^{7,12} average vertical loading rate,^{6,7} instantaneous vertical loading rate,^{7,8,12} and peak braking force.¹³ The vertical impact transient was defined as the minimum slope of the vertical ground reaction force-time curve during the first 25% of stance.³³ Average and instantaneous vertical loading rate were calculated as the average and

peak slopes, respectively, during the middle 60% of the period between initial contact and the vertical impact transient. Peak braking force was defined as the maximum posterior force observed from initial contact to 50% of stance and is reported as a positive value herein. Kinematic outcomes included step length; hip flexion angle at initial contact; the horizontal distance from the heel to centre of mass at initial contact; shank angle at initial contact; and foot strike angle. Kinematic variable selection was based on ease of measurement in a clinical setting and on potential use for running assessment and retraining.

Statistical analysis: Descriptive statistics (means and standard deviations) were calculated for all demographic variables. We used scatterplots and Pearson correlation coefficients to examine the bivariate relationships among baseline preferred speed, kinematic variables, and kinetic outcomes. Stepwise multiple linear regression was used to create forced entry models evaluating the amount of variance in each kinetic outcome explained by baseline preferred speed and the five kinematic variables. Speed was entered into the model first to control for its effect on kinetic outcomes.^{7,34} The kinematic variable with the highest significant partial correlation coefficient was then added to the equation, repeating this process until the addition of more variables did not significantly improve the prediction accuracy of the model.³⁵ Regression diagnostics were conducted on all models using residual analysis to ensure that the equations satisfied the assumptions for linear modelling. All statistical analyses were performed using IBM SPSS Statistics for Windows, Version 22.0 (IBM Corp., Armonk, USA).

Results

Seventy-four female participants met the inclusion criteria and underwent the baseline biomechanical assessment (Table 1). Two participants were excluded based on the absence of a flight phase and one was excluded due to data collection errors. As a result, seventy-one

participants (mean age 36.8 (SD 8.3) years; BMI 22.9 (SD 2.5) kg/m²; baseline preferred speed 2.45 (SD 0.33) m/s; running experience 8.0 (SD 7.4) years) were included in the final analysis.

Significant Pearson product moment correlations were found between kinetic outcomes, baseline preferred speed, and kinematic predictors (Table 2). Peak braking force had a strong positive correlation with step length, and weak positive correlations with heel to centre of mass distance and hip flexion angle—indicating higher braking forces with longer step lengths, greater heel to centre of mass distance, and increased hip flexion at initial contact. Average vertical loading rate and instantaneous vertical loading rate had weak positive correlations with step length, with instantaneous vertical loading rate also having a weak positive correlation with hip flexion angle—indicating greater vertical loading rates with longer step lengths and increased hip flexion at initial contact. Vertical impact transient had weak negative correlations with heel to centre of mass distance and shank angle—indicating higher vertical impact transient magnitude with decreased heel to centre of mass distance and a more vertically-oriented shank. All kinetic outcomes had weak to moderate positive correlations with baseline preferred speed, indicating higher magnitudes and rates with increased speed.

The total amount of variance in the kinetic outcomes explained by the speed and selected kinematic variables ranged from 37-54% (Table 3). A model including speed, step length, foot strike angle, and heel to centre of mass distance explained 54% of variance in peak braking force ($p=.022$). A longer step length and a greater (more rearfoot) foot strike angle were also associated with a greater peak braking force. When step length and foot strike angle were accounted for, a decreased heel to centre of mass distance was also associated with a greater peak braking force.

A model including speed, foot strike angle, and heel to centre of mass distance explained 37% of variance in average vertical loading rate ($p=.001$) and 49% variance in instantaneous

vertical loading rate ($p < .001$). A greater foot strike angle also predicted greater average and instantaneous vertical loading rates. Similarly to peak braking force, after accounting for foot strike angle, increased average and instantaneous vertical loading rates were associated with a decreased heel to centre of mass distance.

A model including speed and heel to centre of mass distance explained 46% of variance in vertical impact transient ($p < .001$). After accounting for baseline preferred speed, a moderate amount of the variance in the magnitude of the vertical impact transient was explained by heel to centre of mass distance, with shorter distances associated with higher magnitudes.

Excessive multicollinearity was not observed in any model (Table 4; variance inflation factor < 10). Residual analyses indicated that all data were consistent with the assumptions of linear regression.

Discussion

The purpose of this study was to examine the contribution of several clinically useful kinematic variables to kinetic outcomes associated with running-related injury. Our findings indicate that a moderate amount (37-54%) of variance in kinetic outcomes associated with running-related injury (peak braking force, average vertical loading rate, instantaneous vertical loading rate, and vertical impact transient) can be explained by several discrete variables related to overstriding and foot strike posture. Although the multivariable relationship between some kinematic variables and kinetic outcomes has been described previously,^{22,23,25,36} we are unaware of any studies that have quantified the multivariable associations between the selected kinetic outcomes associated with running-related injury and clinically-meaningful kinematic measures of overstriding. While increased step length²⁵ and heel to centre of mass distance^{23,25} have been previously associated with higher vertical loading rates and braking impulse, oft-used clinical

measures such as hip flexion angle or shank angle at initial contact have not been related to kinetic outcomes associated with running-related injury risk. Moreover, the relationship between running kinematics and peak braking force—an important risk factor of running-related injury based on a recent study¹³—has not previously been investigated. Findings from the current study address these gaps.

While speed was a significant contributor to all kinetic outcome models, it explained more variance in the peak braking force model than in the vertical ground reaction force models. Running at a faster speed requires a relatively greater propulsive force. Therefore, given the equal and opposite relationship of braking and propulsive impulse during constant speed running on a treadmill, it is not surprising that speed was the greatest factor in explaining the variance in peak braking force. Our findings indicate that short of decreasing speed—an undesired intervention in most cases—peak braking force may also be reduced by shortening step length and transitioning away from a rearfoot strike pattern. Importantly, peak braking force was linked to a significantly higher risk of running-related injury in a recent prospective investigation.¹³

After controlling for speed, step length, foot strike angle, and heel to centre of mass distance accounted for the remaining variance in the peak braking force model. Shank angle and hip flexion angle were not significant contributors to the explained variance when accounting for other variables. While a reduction in peak braking force from a decreased step length has not been reported previously, the likely mechanism for reducing peak braking force is via a reorientation of the ground reaction force vector, resulting in a less posteriorly oriented component. Using the unstandardized regression coefficient values (Table 4), we can determine the change in peak braking force per unit of each kinematic variable. For instance, a change of 1 unit (1 m) in step length is equal to a change of 0.267 BW in peak braking force. Therefore, after accounting for

speed, to achieve a 0.04 BW reduction in peak braking force—a magnitude of one standard deviation in this sample (Table 1) and an 8-fold difference in injury risk based on the results from the previously mentioned study¹³—would require a decrease of 0.148 m in step length (or 16.6%). Some studies have suggested a decrease of only 7.5-10% in step length is necessary to reduce vertical loading rates meaningfully.^{21,37} Therefore the clinical feasibility of decreasing step length by the amount required in our study is questionable. Similarly, while a more forefoot strike angle corresponded to a decreased peak braking force, the clinical meaningfulness of this finding is debatable given that a decrease of 20° in foot strike angle would be required to decrease peak braking force by one standard deviation (0.04 BW), for a given speed and step length.

In our Pearson correlation analysis, heel to centre of mass distance was positively correlated with peak braking force, indicating higher peak braking force with greater heel to centre of mass distance. However, once speed, step length, and foot strike angle were accounted for, greater distances between the heel and centre of mass were actually associated with decreased braking forces. The direction of this relationship is in contrast to our hypothesis and to findings by two other studies.^{21,22} Lieberman et al²² posited that “the braking effects of ‘overstriding’ result from the position of the foot at landing relative to the body’s centre of mass rather than to the position of the foot relative to the knee.” Further, in a study that varied step frequency, Heiderscheit et al²¹ reported that an increase in heel to centre of mass distance was associated with an increase in braking forces.

When viewed in isolation, the relationship between heel to centre of mass distance and peak braking force reported in the previous two studies and seen in our Pearson correlation analysis is logical considering the increased posterior component of the ground reaction force vector with increased heel to centre of mass distance. However, in our regression analysis, when either speed

or step length was added to the peak braking force model, heel to centre of mass distance was not a statistically significant contributor to the variance. Only after foot strike angle was added to the model with speed and step length did heel to centre of mass distance contribute a statistically significant share of the variance. Indeed, speed and step length made up almost 50% of the variance in this model, regardless of the order of entry. Therefore, once these variables have been accounted for, the further information gained from heel to centre of mass distance may have statistical significance, but is not clinically meaningful. In further support of this interpretation, our regression analysis revealed that after accounting for speed, step length, and foot strike angle, a 10 cm increase in heel to centre of mass distance would be required to decrease peak braking force by one standard deviation (0.04 BW).

Vertical loading rate—whether average or instantaneous—has been associated with running-related injury in a number of retrospective studies^{5,7,8} and two systematic reviews.^{6,11} This association has been less consistent from prospective studies. Two large prospective studies^{9,10} that reported an association between high loading rates and running-related injury compared injured vs. uninjured runners—instead of high vs. low loading rate—which may limit the scope of conclusions from such prospective cohort studies. Two other prospective studies found no association between loading rate and running-related injury among female¹³ or collegiate runners.³⁸ The average and instantaneous vertical loading rate models in this study consisted of the same subset of kinematic variables to explain 37% and 49% of the variance of those kinetic outcomes, respectively. After controlling for speed, foot strike angle and heel to centre of mass distance accounted for the remaining variance in those models. A more rearfoot strike angle was associated with increased average and instantaneous vertical loading rates. The significant contribution of foot strike angle to the variance in average and instantaneous vertical loading rates

fits with previous work.²³ The lower vertical loading rates seen with a more forefoot strike pattern might be explained by a functionally longer lever of the lower extremity at initial contact, producing a larger arc in the vertical trajectory of the centre of mass.³⁹ Our results suggest that moving from a rearfoot strike to a more forefoot strike could be an appropriate gait retraining intervention to reduce vertical loading rate. Indeed, studies that have converted runners from a rearfoot to a forefoot strike pattern have shown a reduction in vertical loading rates.^{28,39} However, the regression coefficient in our analysis suggests that a decrease in foot strike angle of 14.5° would be necessary to decrease the average vertical loading rate by one standard deviation (11.9 BW/s). This may not be clinically feasible in many cases.

The Pearson correlation analysis demonstrated no significant relationship between heel to centre of mass distance and average or instantaneous vertical loading rate. However, a decreased heel to centre of mass distance was associated with greater average and instantaneous vertical loading rates once accounting for speed and foot strike angle. Considering the increased vertical component of the ground reaction force vector when there is a decreased heel to centre of mass distance, a negative relationship between vertical loading rate and heel to centre of mass distance is logical. This is in agreement with a recently published study by Harrison et al,²⁵ which found that heel to centre of mass distance was a predictor of vertical ground reaction force variables and maintained the same inverse relationship. Heel to centre of mass distance explains a greater amount of the variance in the average and instantaneous vertical loading rate models than in the peak braking force model (12% and 19% versus 4%, respectively) after accounting for other variables. Again, the clinical practicality of this relationship is questionable given that a one standard deviation decrease in average or instantaneous vertical loading rate would require an increase of

6.9 cm or 5.4 cm in heel to centre of mass distance for a given speed and foot strike angle, respectively.

Heel to centre of mass distance accounted for a substantial amount (39%) of the variance in the vertical impact transient model after controlling for speed. No other kinematic variable contributed to the model. Once again, higher magnitudes were associated with a decreased heel to centre of mass distance. This is counter to findings by Heiderscheit et al²¹ who found greater heel to centre of mass distance corresponded to increased frequency of vertical impact transient events, although magnitudes were not reported. Lieberman et al²² reported that higher vertical impact transient magnitudes were associated with a more extended lower extremity and greater horizontal distance between ankle joint centre and knee joint centre. While the authors did not report the association between vertical impact transient magnitudes and horizontal distance between ankle joint centre and centre of mass (a measure more comparable to heel to centre of mass distance in our study), one would expect that the positive relationship would persist. Given the more vertically oriented ground reaction force vector at decreased heel to centre of mass distance, the negative relationship between vertical impact transient and heel to centre of mass distance in our Pearson correlation analysis and regression model is reasonable. The large contribution of heel to centre of mass distance to the vertical impact transient model is supported by the more clinically meaningful regression coefficient, which predicts that an increase of 4.2 cm in heel to centre of mass distance would result in a decrease of one standard deviation of vertical impact transient (0.25 BW), after accounting for speed.

Speed, step length, foot strike angle, and heel to centre of mass distance were significant contributors to our kinetic outcome models. However, hip flexion angle and shank angle did not contribute significantly to any of the models, indicating that these variables are of limited value

relative to the other kinematic variables when estimating kinetic outcomes. While shank angle is often used clinically to estimate the degree of overstriding, it is closely linked to the angle of knee flexion at initial contact, which has been shown to have no significant association with kinetic outcomes in two previous studies.^{22,23} Shank angle also had a moderate correlation with heel to centre of mass distance, likely due to the increased heel to centre of mass distance when the shank is angled forward from the vertical plane. Therefore, after accounting for heel to centre of mass distance in the models, shank angle likely did not provide any further information to explain the variance.

There are some limitations to this study. First, our sample includes females only. However, Wille et al²³ found that sex did not directly affect the relationship between kinematic variables and kinetic outcomes. Therefore, we are confident that our results can also be generalized to male runners. Second, our analysis was conducted on a treadmill and while kinetics have been shown to be similar between treadmill and overground running,^{40,41} some differences in kinematics have been documented (decreased step length and heel to centre of mass distance;⁴² more flexed knee at initial contact;⁴² more midfoot/forefoot strike^{42,43}), which may limit the generalizability of the results to overground running. Finally, only 18-39% of the variance in kinetic outcomes was accounted for by the variables of interest after controlling for speed. Therefore, there are likely other factors that influence vertical and anterior-posterior ground reaction force during the loading phase of running that were not included in this analysis. Importantly, we reported step length instead of step frequency in this study. This was based on work by Mercer et al²⁴ that showed that increases in step length, rather than decreases in step frequency, decreased shock attenuation in runners. However, we acknowledge that step rate is a more accessible measure to use in a clinical

context, though this may be changing with the improvement and availability of wearable technology.^{44,45}

Debate in the literature and among clinicians exists as to how runners should be instructed to achieve a reduction in kinetic variables associated with injury, with experts disagreeing on the importance of reducing overstride versus changing foot strike pattern.¹⁴ Some experts suggest that reducing overstride may be beneficial for reducing running-related injury risk.¹⁴ However, there is no consensus on what constitutes overstriding and there has been little evidence to support the importance of overstriding to running-related injury risk. While a definition of what constitutes an overstride is important, we suggest that the kinetic consequence of that overstride is more important. While a reduction in speed is one option for decreasing vertical and anterior-posterior ground reaction forces, for runners wanting to maintain their preferred speed, several kinematic modifications may also achieve these outcomes. Our findings suggest that to reduce peak braking force, a focus on shortening step length and transitioning away from a rearfoot strike pattern may be appropriate strategies. Further, to reduce vertical loading rates, we suggest that moving from a rearfoot strike to a more forefoot strike might be an appropriate gait retraining intervention. Modifying shank angle and hip flexion angle do not appear to significantly contribute to the variance of these kinetic outcomes. Furthermore, heel to centre of mass distance—often cited as a measure of overstriding—appears to have little to contribute to a reduction in peak braking force once speed, step length, and foot strike angle are accounted for. While these results contribute to our understanding of the explained variance of kinetic outcomes associated with running-related injury, the necessary magnitude of changes in kinematics reported in this study may be too great for any one modification in isolation to achieve the desired result. An interventional study aimed at modifying some of the kinematic parameters identified in this study is necessary to determine

which of these variables are best able to reduce kinetic outcomes clinically. Further prospective investigation into the strategies employed by runners to reduce vertical and anterior-posterior ground reaction forces would also be valuable.

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Figure 1 – Marker positions for static trial.

Table 1. Kinetic and kinematic running gait data of the participants (n=71).

	Mean (SD)	Min, max
PBF (BW)	0.253 (0.041)	<u>0.167</u> , 0.362
AVLR (BW/s)	45.52 (11.89)	21.33, 70.24
IVLR (BW/s)	57.49 (15.47)	27.46, 92.59
VIT (BW)	1.48 (0.25)	0.95, 2.17
Step Length (m)	0.89 (0.10)	0.68, 1.15
Foot strike angle (°)	4.34 (7.05)	-21.04, 16.70
Horizontal Distance from Heel to COM (m)	0.102 (0.030)	0.02, 0.18
Shank angle (°)	5.96 (2.58)	-0.05, 10.23
Hip flexion angle at initial contact (°)	30.96 (5.79)	18.19, 46.75

PBF, peak braking force; AVLR, average vertical loading rate; IVLR, instantaneous vertical loading rate; VIT, vertical impact transient. Foot strike angle measured from the horizontal with a positive angle a more rearfoot strike. Shank angle measured from the vertical with a positive angle indicating that the ankle is anterior to the knee.

Table 2. Pearson product moment correlations (95% confidence interval) among kinetic risk factors, baseline preferred speed, and kinematic variables.

	Baseline Preferred Speed	Step Length	Foot strike angle	Horizontal distance from heel to COM	Shank angle	Hip flexion angle at initial contact
SL	0.88* (0.81, 0.92)					
FSA	-0.10 (-0.32, 0.14)	-0.12 (-0.34, 0.12)				
HCOM	0.49* (0.29, 0.65)	0.57* (0.39, 0.71)	0.34* (0.11, 0.53)			
SA	0.22 (-0.02, 0.43)	0.26* (0.03, 0.47)	-0.00 (-0.24, 0.23)	0.54* (0.35, 0.69)		
HFA	0.37* (0.15, 0.55)	0.32* (0.09, 0.51)	0.16 (-0.08, 0.38)	0.16 (-0.07, 0.38)	-0.10 (-0.32, 0.14)	
PBF	0.67* (0.51, 0.78)	0.70* (0.55, 0.80)	0.10 (-0.14, 0.32)	0.33* (0.11, 0.53)	0.03 (-0.20, 0.26)	0.36* (0.14, 0.55)
AVLR	0.42* (0.21, 0.60)	0.35* (0.13, 0.54)	0.27* (0.04, 0.47)	0.06 (-0.17, 0.29)	0.05 (-0.19, 0.28)	0.27* (0.04, 0.47)
IVLR	0.48* (0.28, 0.64)	0.36* (0.13, 0.54)	0.25* (0.02, 0.46)	0.01 (-0.22, 0.24)	-0.04 (-0.27, 0.20)	0.31* (0.08, 0.50)
VIT	0.30* (0.07, 0.50)	0.23 (-0.00, 0.44)	-0.22 (-0.43, 0.01)	-0.40* (-0.58, -0.18)	-0.32* (-0.51, -0.09)	0.13 (-0.11, 0.35)

*Denotes significantly different from zero at $p < .05$.

SL, step length; FSA, foot strike angle; HCOM, horizontal distance from heel to COM at IC; SA, shank angle; HFA, hip flexion angle; PBF, peak braking force; AVLR, average vertical loading rate; IVLR, instantaneous vertical loading rate; VIT, vertical impact transient.

Table 3. Summary of regression models for predicting kinetic risk factors.

	R ²	Adjusted R ²	R ² Change	p
<u>Peak braking force</u>				
Speed	0.45	0.44	0.45	<.001
Speed + SL	0.50	0.49	0.05	.009
Speed + SL + FSA	0.53	0.51	0.03	.036
Speed + SL + FSA + HCOM	0.57	0.54	0.04	.022
<u>Average vertical loading rate</u>				
Speed	0.18	0.17	0.18	<.001
Speed + FSA	0.28	0.26	0.10	.003
Speed + FSA + HCOM	0.40	0.37	0.12	.001
<u>Instantaneous vertical loading rate</u>				
Speed	0.23	0.22	0.23	<.001
Speed + FSA	0.32	0.30	0.09	.004
Speed + FSA + HCOM	0.52	0.49	0.19	<.001
<u>Vertical impact transient</u>				
Speed	0.09	0.08	0.09	.012
Speed + HCOM	0.48	0.46	0.39	<.001

Speed, baseline preferred speed; SL, step length; HCOM, horizontal distance from heel to COM; FSA, foot strike angle.

Table 4. Unstandardized regression coefficients and variance inflation factors for variables in models.

	B (SE)	VIF
<u>Peak braking force (BW)</u>		
Constant	−0.026 (0.031)	−
Speed	0.029 (0.021)	4.33
Step length	0.267 (0.073)	5.22
Foot strike angle	0.002 (0.001)	1.34
Heel to COM	−0.358 (0.153)	1.97
<u>Average vertical loading rate (BW/s)</u>		
Constant	−1.98 (8.77)	−
Speed	25.12 (4.20)	1.47
Foot strike angle	0.82 (0.18)	1.26
Heel to COM	−172.42 (47.60)	1.64
<u>Instantaneous vertical loading rate (BW/s)</u>		
Constant	−12.37 (10.25)	−
Speed	38.42 (4.90)	1.47
Foot strike angle	1.14 (0.21)	1.26
Heel to COM	−286.17 (55.61)	1.64
<u>Vertical impact transient (BW)</u>		
Constant	0.87(0.16)	−
Speed	0.49 (0.08)	1.32
Heel to COM	−5.81 (0.82)	1.32

B, unstandardized regression coefficient; SE, standard error; Speed, baseline preferred speed; VIF, variance inflation factor. The B values are expressed in units of kinetic outcome variable for every per unit increase in the predictor variables (speed, m/s; step length, m; foot strike angle, °; heel to COM, m).